

## Set notation and sample space



1 Heads and Tails

2

a  $\{1, 2, 3, 4, 5, 6\}$

b  $\{1, 2\}$

c  $\{3, 6\}$

d  $\{2, 4, 6\}$

3

a 1, 2, 3, 4, 5, 6

b 1, 2

4

a  $\{2, 4, 6, 8, 10, 12, 14\}$

b  $\{-7, -6, -5, -4\}$

c  $\{1, 3, 5, 7, 9, 11\}$

d  $\left\{1, \frac{1}{10}, \frac{1}{100}, \frac{1}{1000}, \frac{1}{10\,000}, \frac{1}{100\,000}\right\}$

e  $\{2, 4, 8, 16, 32, 64\}$

5

a  $\{1, 2, 4, 8, 16\}$

b  $\{1, 2, 4, 8\}$

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## Set notation (Extended)

6  $\{x : x \text{ is an even number and } 2 \leq x \leq 8\}$

7

a True

b True

c False

d True

8

a  $\subseteq$

b  $\subseteq$

c  $\not\subseteq$

d  $\not\subseteq$

9

a  $\in$

b  $\notin$

10

a  $\{g, a, l, o, p\}$

b Yes

c  $\{p, a, r, l, e, o, g, m\}$

11 The set of all the elements in the universal set that are not in set  $A$ .

12

a  $A = \{0, 1, 2, 3, 4, 5\}$



b Yes

13

a 11

b 12

c 7

d 16

14

a No

b Yes

c No

d Yes

15

a Because there is only one point that satisfies both equations, and it occurs at the point of intersection of the lines.

b (10, 26)

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## Operations on sets

16

a The set containing all the elements that are members of set  $A$  or of set  $B$  or of both sets.

b The set containing all the elements that are common to both set  $A$  and set  $B$ .

17

a Yes

b {cauliflowers}

18 {3, 7}

19

a {0, 4, 16, 36, 64}

b {5, 10, 15, 20, 25, 30, 35, 40, 45}

20

a {1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36}

b {1, 2, 3, 4, 6, 12}

21

a {1, 2, 3, 4, 5, 6}

b {1, 2, 3, 4, 5, 6, 7, 8}

c {1, 2, 3, 4, 5, 6}

d {1, 2, 3, 4, 5, 6, 7, 8}

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## Complement of sets (Extended)

22

- a
- i {roses, daisies}      ii {lillies}
- iii {roses, daisies, lillies, sunflowers}      iv {roses, daisies}
- b Yes



23

- a Yes      b 5
- c {7, 13, 12, 26}      d  $U = \{7, 8, 9, 12, 13, 21, 26, 28, 30\}$
- e The empty set  $\emptyset$ .

24

- a {5, 6, 7}      b  $\emptyset$

25

- a People who do not like softball.      b People who like either football, softball, or swimming.

26

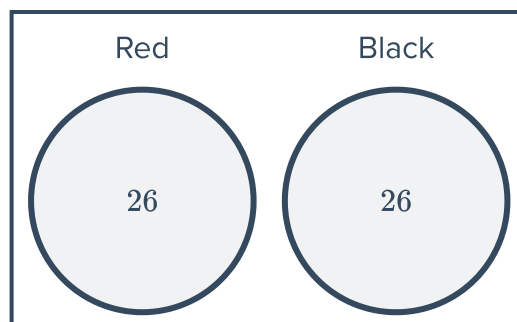
- a Yes      b No      c  $\{-5, 3\}$       d {11}
- e 19      f 16

## Venn diagrams

27

- a Venn diagram 4      b Venn diagram 3
- c Venn diagram 2      d Venn diagram 1

28



29

- a  $A \cap B = \{7, 8\}$       b  $A \cup B = \{2, 4, 6, 7, 8, 10, 14\}$



## Venn diagrams (Extended)

30

a  $A' = \{7, 14, 12, 10\}$

b  $B' = \{4, 5, 1, 12, 10\}$

31

a  $A = \{1, 3, 4, 5, 9, 10\}$

b  $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

c  $B' = \{1, 2, 9, 10, 11, 13\}$

32

a  $A \cap B' = \{3, 4, 6\}$

b  $(A \cup B)' = \{15, 60\}$

33

a Yes

b Yes

34

a  $\{4, 5, 9\}$

b  $\{1, 2, 3, 7, 9, 10, 11, 12, 13, 14\}$

c  $\{4, 5\}$

d  $\{3, 4, 5, 9\}$

e  $\{1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

35

a Yes

b No

36

a 2

b 5

c 3

d 4

e 8

f 7

g 1

h 6

37

a

|                        | Play Rugby League | Don't play Rugby League |
|------------------------|-------------------|-------------------------|
| Play Rugby Union       | 43                | 185                     |
| Don't play Rugby Union | 179               | 146                     |

b

i 179

ii 222

iii 228

iv 185

v 331

vi 325